

13-3b Average and Marginal Cost

As the owner of his firm, Caleb has to decide how much to produce. When making this decision, he will want to consider how the level of production affects his firm's costs. Caleb might ask his production supervisor the following two questions about the cost of producing coffee:

- How much does it cost to make the typical cup of coffee?
- How much does it cost to increase production of coffee by 1 cup?

These two questions might seem to have the same answer, but they do not. Both answers are important for understanding how firms make production decisions.

To find the cost of the typical unit produced, we divide the firm's costs by the quantity of output it produces. For example, if the firm produces 2 cups of coffee per hour, its total cost is \$3.80, and the cost of the typical cup is \$3.80/2, or \$1.90. Total cost divided by the quantity of output is called average total cost (total cost divided by the quantity of output). Because total cost is the sum of fixed and variable costs, average total cost can be expressed as the sum of average fixed cost and average variable cost.

Average fixed cost (fixed cost divided by the quantity of output) equals the fixed cost divided by the quantity of output, and

average variable cost (variable cost divided by the quantity of output) equals the variable cost divided by the quantity of output.

Average total cost tells us the cost of the typical unit, but it does not tell us how much total cost will change as the firm alters its level of production. Column (8) in Table 2 shows the amount that total cost rises when the firm increases production by 1 unit of output. This number is called

marginal cost (the increase in total cost that arises from an extra unit of production). For example, if Caleb increases production from 2 to 3 cups, total cost rises from \$3.80 to \$4.50, so the marginal cost of the third cup of coffee is \$4.50 minus \$3.80, or \$0.70. In the table, the marginal cost appears halfway between any two rows because it represents the change in total cost as quantity of output increases from one level to another.

It is helpful to express these definitions mathematically:

$$\begin{aligned}\text{Average total cost} &= \text{Total cost}/\text{Quantity} \\ ATC &= TC/Q,\end{aligned}$$

and

Marginal cost = Change in total cost/Change in quantity

$$MC = \Delta TC / \Delta Q.$$

Here Δ , the Greek letter delta, represents the change in a variable. These equations show how average total cost and marginal cost are derived from total cost. *Average total cost tells us the cost of a typical unit of output if total cost is divided evenly over all the units produced. Marginal cost tells us the increase in total cost that arises from producing an additional unit of output.* In the next chapter, business managers like Caleb need to keep in mind the concepts of average total cost and marginal cost when deciding how much of their product to supply to the market.

Chapter 13: The Costs of Production: 13-3b Average and Marginal Cost

Book Title: WGU Economics

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